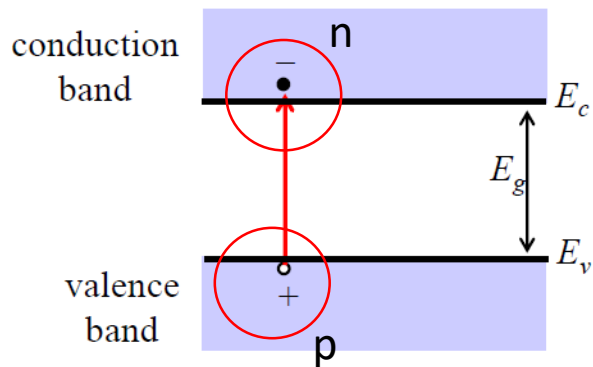


Intrinsic Semiconductor

No doping

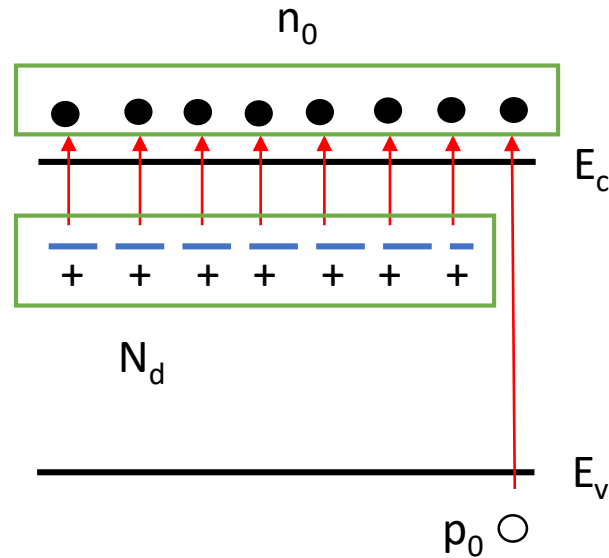


At thermal equilibrium

$$n_0 = p_0 = n_i$$

n-type

Doped with group V materials
(P, As, Sb)



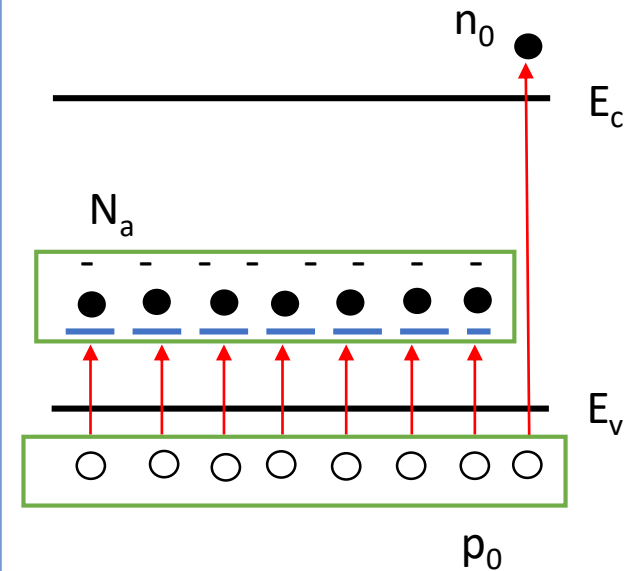
At $k_B T \gg$ ionization energy

$$n_0 = N_d + p_0$$

$$n_0 > p_0$$

p-type

Doped with group III materials
(B, Al, Ga)



At $k_B T \gg$ ionization energy

$$p_0 = N_a + n_0$$

$$p_0 > n_0$$